

ECN 711
Problem Set #1

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Due date: Saturday, September 6th, 2025.

1. Consider the Solow model.

1. Show that the savings rate that achieves the golden rule capital to effective labor ratio, s_{GR} , equals the share of capital in total income.
2. From now on, assume that the aggregate technology is Cobb-Douglas: $Y = K^\alpha (AL)^{1-\alpha}$. Suppose $k(0) < k^*$. Show that the growth rate for output per-capita declines as the economy converges to the steady state.
3. Suppose the economy starts with $k_0 < k^*$, where k^* denotes the steady state capital per-effective labor. What is the associated dynamics for the share of capital as the economy converges to the steady state? Explain carefully.
4. Consider an economy along a balanced growth path described by the following parameters: $\delta = 0.06$, $n = 0.01$, $g = 0.02$ and $\alpha = 1/3$. Find the saving rate so that, along the balanced growth path, the *capital to output ratio* equals 3.0.

2. Consider the dynamics of capital accumulation in the following simple version of the Solow model. Each member of the labor force is endowed with one unit of productive time, of which all of them devote a fixed fraction h to work. Population grows at the rate n , and the total size of the population at time t is N_t . Output is produced through a constant returns to scale technology that requires both capital and labor. Capital depreciates at the rate $\delta \in (0, 1]$. For simplicity, assume that there is no growth in labor efficiency.

1. Define aggregate labor, L_t . Derive the law of motion for the capital to labor ratio.

2. Analyze qualitatively the dynamic effects on the capital to labor ratio of a sudden and permanent increase in the fraction of time devoted to work, h .
3. Suppose labor is paid its marginal product. Describe the dynamics of the wage rate following the increase in h . Explain.
4. Is output *per-capita* higher in the new steady state? Explain.

3. Consider now the following version of the Solow Model, where the production technology requires two distinct types of capital, K_1 and K_2 . The single good in the economy is produced via

$$Y = \bar{A} K_1^\alpha K_2^\beta (L)^{1-\alpha-\beta}.$$

Suppose for simplicity that there is no growth in labor efficiency. Population grows at the rate n . The parameter $\bar{A} > 0$ stands for the level of Total Factor Productivity.

As in the standard Solow model, there are constant saving rates for both types of capital. Let s_1 and s_2 be the exogenous saving rates for capital of types 1 and 2. Both types of capital depreciate at the same rate, $\delta \in (0, 1]$.

1. Derive the laws of motion for capital per worker of both types.
2. Solve explicitly for the steady-state values of capital per worker of both types.
3. Solve for the steady-state level of output per worker.

What is the *elasticity* of output per worker in steady state with respect to changes in Total Factor Productivity? How does this elasticity compare to the standard case with one type of capital? Explain.

4. This exercise asks you to address the issue of whether a balanced growth path is possible in the Solow model whenever a factor of production is in fixed supply (*land* for example). Suppose the technology displays constant returns to scale and has the Cobb Douglas form:

$$Y = K^{\alpha_1} (AL)^{\alpha_2} X^{1-\alpha_1-\alpha_2},$$

where X denotes the available quantity of the factor in fixed supply, with $X_t = \bar{X}$ all t . Population or raw labor (L) and labor efficiency (A) grow at the rates $(1+n)$ and $(1+g)$, respectively.

1. What is the rate of growth of aggregate output along a balanced growth path?
2. Under what conditions there is long run growth in per-capita output? Explain.
3. From now onwards, assume that the conditions that guarantee long-run growth in output per capita hold.

Note that the growth rate derived in the first point implicitly defines the variable needed to construct a normalized capital stock variable, $\tilde{k}$. That is, the variable Z so that $\tilde{k} = K/Z$.

Find the law of motion that governs the dynamics of $\tilde{k}$. Argue that for any $\tilde{k} > 0$, this variable converges monotonically to a steady state value, and thus a unique balanced growth path exists.

4. Describe the effects of an exogenous increase in the stock of the fixed resource, from $\bar{X}_L$ to $\bar{X}_H$. Explain.
5. Evaluate the following assertion: “An economy with a higher stock of a fixed resource will have a higher capital to output ratio in the long run”.
5. Consider the version of the Solow model described in the previous question. Derive the condition characterizing the golden rule of capital accumulation. Explain.
6. Consider the C.E.S. (Constant Elasticity of Substitution) production function:

$$F(K, AL) = [a(K)^\Psi + (1-a)(AL)^\Psi]^{1/\Psi},$$

with $a \in (0, 1)$ and $\Psi < 1$. This production function displays constant *elasticity of substitution* between capital and labor equal to $1/(1 - \Psi)$.

1. Show that the CES production function displays constant returns to scale. Obtain the production function in intensive form, $f(k)$.
2. Obtain $f'(k)$. Show that in each of the cases (i) $\Psi < 0$ and (ii) $0 < \Psi < 1$, one of the limiting Inada conditions is not satisfied.
3. Show that as $\psi \rightarrow 0$, the production function converges to the Cobb-Douglas function:

$$\lim_{\Psi \rightarrow 0} F(K, AL) = K^a (AL)^{1-a}$$

(Hint: consider the limit of $\log[F(K, AL)]$.)